S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 11
Postlab (OPEN ENDED PRACTICAL)

Aim: To implement the First Fit Memory Allocation Algorithm and allocate memory blocks to processes.

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❖ **Aim:** To implement the First Fit Memory Allocation Algorithm and allocate memory blocks to processes.

❖ **Objectives:**

- To understand memory management in operating systems
- To implement First Fit allocation technique
- To allocate processes to available memory blocks
- To analyze memory utilization

❖ **Requirements:**

✓ **Hardware Requirements:**

- Computer system (PC/Laptop)
- Minimum 2 GB RAM
- Keyboard and monitor

✓ **Software Requirements:**

- C Compiler (Turbo C / GCC / Code:Blocks)
- Operating System (Windows/Linux)

❖ **Theory:**

Memory allocation is a process of assigning available memory blocks to processes.
In **First Fit Algorithm**:

- The first available memory block that is large enough is allocated to the process
- Once allocated, that block is reduced or marked as used
- It is simple and fast but may lead to fragmentation

✓ **Key Points:**

- Easy to implement
- Faster than other allocation methods
- May cause internal fragmentation

❖ **Algorithm**

1. Input number of memory blocks and their sizes
2. Input number of processes and their sizes
3. For each process:
    a. Find the first block that can fit
    b. Allocate the block
4. If no block is found → process not allocated
5. Display allocation result

❖ CODE:

```
#include <stdio.h>

int main() {
    int nb, np, i, j;

    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);

    int block[nb];

    printf("Enter sizes of memory blocks:\n");
    for(i = 0; i < nb; i++) {
        scanf("%d", &block[i]);
    }

    printf("Enter number of processes: ");
    scanf("%d", &np);

    int process[np];

    printf("Enter sizes of processes:\n");
    for(i = 0; i < np; i++) {
        scanf("%d", &process[i]);
    }

    int allocation[np];
    for(i = 0; i < np; i++) {
        allocation[i] = -1;
    }

    for(i = 0; i < np; i++) {
        for(j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                allocation[i] = j;
                block[j] -= process[i];
                break;
            }
        }
    }
    printf("\nProcess No.\tProcess Size\tBlock No.\n");

    for(i = 0; i < np; i++) {
        printf("%d\t\t%d\t\t", i+1, process[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
```

Output:

```
Enter number of memory blocks: 4
Enter sizes of memory blocks:
100 500 200 300
Enter number of processes: 4
Enter sizes of processes:
212 417 112 426

Process No.     Process Size    Block No.
1               212             2
2               417             Not Allocated
3               112             2
4               426             Not Allocated


...Program finished with exit code 0
Press ENTER to exit console.
```

❖ **Conclusion**: The First Fit algorithm is simple and fast for memory allocation. However, it may lead to memory fragmentation and inefficient utilization of memory.

❖ **Discussion Questions:**

1. What is First Fit Memory Allocation?
2. What is fragmentation?
3. What are advantages of First Fit?
4. What are disadvantages of First Fit?

❖ **References:**

https://www.geeksforgeeks.org/disk-scheduling-algorithms/
https://chatgpt.com/

Date:____/____/2026

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